

TeSA Cohort 6 Specialization

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3. Part A

3.1 FRD (Functional Requirements Document)

Functional Requirements Document

Trigger → Observable Response | First Release (May Sitting)

Audience: engineers who implement and testers who verify. Each requirement is independently testable.

0. Convention

Every requirement is written as: When [trigger], the system [observable response]. A requirement passes if and only if the stated response is observed under the stated trigger. Requirements are numbered FR-XXX. Money is stored and compared as whole kobo (PIN price = 350000). Cross-references use the same numbers.

1. PIN Purchase

FR-001. When a buyer submits a valid payment request for one PIN with a unique client reference that has never been seen before, the system charges the card exactly once for 350000 kobo, creates one PIN with remaining checks = 5, and returns that PIN and the remaining count to the buyer.

FR-002. When a buyer resubmits a payment request that carries the same client reference as a request already processed successfully, the system does not charge the card again, does not create a second PIN, and returns the same PIN and remaining count that were returned for the original request.

FR-003. When a buyer resubmits a payment request with the same client reference as a request that is still in progress (neither success nor failure recorded), the system does not initiate a second charge and returns a response that indicates the original request is still being processed.

FR-004. When the payment partner returns a decline or failure for a new client reference, the system creates no PIN, records the failure against that client reference, and returns a failure message that does not contain a PIN.

FR-005. When a reconciliation job runs against the payment-partner settlement file, every successful charge in the settlement file has exactly one corresponding PIN in the system, and every PIN in the system has exactly one corresponding successful charge.

FR-006. When a PIN is created, the system stores the PIN value, the remaining check count (initially 5), the creation timestamp, and the client reference that authorised it, and makes the remaining count readable by subsequent check requests that present that PIN.

FR-007. When a buyer submits a payment request that omits a client reference, the system refuses the request, does not charge the card, and returns a message that a client reference is required.

FR-008. When a PIN has been created and the buyer later requests the status of that PIN using the original client reference, the system returns the PIN value and the current remaining check count without creating a new charge or a new PIN.

2. Candidate Result Check

FR-010. When a candidate submits a PIN that exists, has remaining checks greater than zero, and is linked to a candidate who has released results, the system returns the candidate's nine subject grades, decrements the remaining check count by one, and returns the new remaining count.

FR-011. When a candidate submits a PIN that has already been used five times (remaining count = 0), the system refuses the check, leaves the PIN's remaining count at zero, and returns the message that the PIN is exhausted.

FR-012. When a candidate submits a PIN that does not exist in the system, the system returns a message that the PIN is invalid and does not decrement any remaining count.

FR-013. When a candidate submits a valid PIN but results for that candidate have not yet been released, the system returns a message that results are not yet available and does not decrement the remaining check count.

FR-014. When a candidate submits a valid PIN for a candidate whose results have been released and later amended, the system returns the current (amended) grades, not the original grades, and decrements the remaining count by one.

FR-015. When two concurrent check requests arrive for the same PIN that has remaining count = 1, the system allows exactly one of them to succeed and return grades, sets remaining count to 0, and refuses the other with the exhausted-PIN message.

FR-016. When a successful check is completed, the system records the PIN used, the timestamp, the channel (web / USSD / SMS), and the remaining count after the check, and makes that record available for audit.

FR-017. When a candidate check is performed, the response contains only that candidate's subject grades and the remaining PIN count; it does not contain the candidate's address, telephone number, examination centre, or any other candidate's data.

FR-018. When a candidate submits a PIN that exists but is linked to a different candidate identity than the one implied by any optional identity field in the request, the system refuses the check and returns a message that the PIN does not match the supplied identity, without decrementing the remaining count.

FR-019. When a candidate has used a PIN fewer than five times and later submits the same PIN again, the system returns the current grades (including any amendments made since the previous check) and decrements the remaining count by one.

3. USSD and SMS Channel

FR-020. When a candidate dials the published USSD code and selects the result-check option, the system prompts for a PIN, accepts a PIN entry of the correct length, and applies the same PIN validation and result-return rules defined in FR-010 through FR-019.

FR-021. When a USSD session receives a valid PIN with remaining checks > 0 and released results, the system returns the nine subject grades and the new remaining count inside the USSD response and ends the session without requiring a further network round-trip for the grades themselves.

FR-022. When a USSD session receives a PIN that is exhausted, invalid, or for which results are not yet released, the system returns the corresponding refusal message defined in FR-011, FR-012 or FR-013 inside the USSD response.

FR-023. When a candidate sends an SMS containing the published keyword and a PIN to the published short code, the system validates the PIN under FR-010 through FR-019 and replies with a single SMS containing either the nine subject grades and remaining count, or the appropriate refusal message.

FR-024. When an SMS or USSD check succeeds or fails, the system records the channel as USSD or SMS respectively in the audit record required by FR-016.

FR-025. When a USSD or SMS request is processed, the system completes the response using no more than four network round-trips from the initial dial or SMS send to the final grades or refusal message.

FR-026. When a candidate uses USSD or SMS, the system does not require the candidate to have a smartphone, a data plan, or a browser; the same functional outcomes defined for web checks are available through these channels.

FR-027. When a USSD session times out before the candidate enters a PIN, the system does not create a check record, does not decrement any remaining count, and leaves the PIN state unchanged.

FR-028. When an SMS arrives with the correct keyword but a PIN of incorrect length or non-numeric characters, the system replies with a single SMS stating that the PIN format is invalid and does not attempt a check or decrement any count.

4. School Bulk Download

FR-030. When an authenticated school user requests the bulk result list for their own centre number after the noon release flag has been set, the system returns a complete file containing every candidate entered under that centre for the current sitting, with each candidate's nine subject grades.

FR-031. When an authenticated school user requests the bulk list for a centre number that is not assigned to that school, the system refuses the request and returns a message that the centre is not authorised for that account.

FR-032. When a school requests the bulk list before the noon release flag has been set, the system refuses the request and returns a message that results have not yet been released.

FR-033. When a school bulk file is generated, every grade in that file matches the grade that would be returned by an individual candidate check (FR-010 / FR-014) for the same candidate and subject at the same moment.

FR-034. When a school bulk request is processed, the system serves it from a channel that does not share the request path used by individual candidate checks, so that bulk traffic does not reduce the capacity available for individual checks.

FR-035. When a school bulk file is returned, it contains only candidates belonging to the requesting school's centre; it does not contain candidates from any other centre.

FR-036. When a school bulk download completes, the system records the school account, centre number, timestamp, and number of candidates in the file for audit.

FR-037. When a school whose centre has candidates with amended grades requests the bulk list after those amendments, the system returns the amended grades in the file, consistent with FR-033 and FR-014.

FR-038. When an unauthenticated caller requests a school bulk list, the system refuses the request and returns no candidate data.

5. Verification Request

FR-040. When an authenticated verifier submits a verification request that names a candidate and one or more subjects for which results have been released, the system returns a validity flag of true and the exact current grades for only those named subjects.

FR-041. When an authenticated verifier submits a verification request for a candidate or subject combination that has no released result, the system returns a validity flag of false and does not return any grade values.

FR-042. When a verification response is returned, it contains only the requested subject grades and the validity flag; it does not contain the candidate's other subjects, address, telephone number, examination centre, or any field not required for the verification decision.

FR-043. When a verification request is made for a candidate whose grades have been amended after release, the system returns the current (amended) grades, not the grades that were originally released.

FR-044. When a verification request is submitted without authentication, or with credentials that are not authorised for verification, the system refuses the request and returns no grade data.

FR-045. When a successful verification is completed, the system records the verifier identity, the candidate and subjects requested, the timestamp, and the grades returned, and makes that record available for audit.

FR-046. When a verification request names subjects that have been released and subjects that have not, the system returns grades only for the released subjects, a validity flag that reflects only those subjects, and does not invent grades for the unreleased subjects.

FR-047. When a verifier submits a request that names a candidate but omits every subject, the system refuses the request and returns a message that at least one subject must be specified.

FR-048. When two verifiers independently request the same candidate and subject at the same time, each receives an identical validity flag and grade set, and each request produces its own audit record under FR-045.

6. Amendment Workflow

FR-050. When an authenticated amendment officer submits a grade change for a named candidate and subject together with an evidence reference, the system records the officer identity, the timestamp, the previous grade, the new grade, and the evidence reference, and makes the new grade the value returned by all subsequent candidate checks and verification requests for that subject.

FR-051. When an amendment is accepted, the system makes the new grade visible to a subsequent candidate check (FR-010 / FR-014) and to a subsequent verification request (FR-040 / FR-043) within five minutes of the amendment being recorded.

FR-052. When an amendment officer submits a grade change without an evidence reference, or without authentication, the system refuses the amendment and does not change any stored grade.

FR-053. When an amendment has been recorded, the system does not alter or delete the audit record of that amendment; the record remains queryable for the lifetime of the result archive.

FR-054. When a candidate check or verification is performed after an amendment, the system returns the amended grade and does not return the pre-amendment grade as the current value.

FR-055. When an amendment is recorded for a candidate whose certificate has already been verified by a third party, the system does not send a recall or invalidation message to that third party; only future checks and verifications reflect the new grade.

FR-056. When a regulator or authorised auditor requests the history of a named candidate's subject result, the system returns the original released grade, every subsequent amendment in chronological order, and for each amendment the officer identity, timestamp and evidence reference.

FR-057. When two amendment requests for the same candidate and subject arrive concurrently, the system applies them in a defined order, leaves a complete audit trail for both, and ensures that the final current grade is exactly one of the submitted values (the later one under the defined order).

FR-058. When an amendment officer submits a grade change for a candidate or subject that has no released result, the system refuses the amendment and returns a message that there is no released result to amend.

FR-059. When an amendment changes a grade and a school later requests the bulk list for that centre, the bulk file contains the amended grade, consistent with FR-033 and FR-037.

7. Cross-Cutting Observable Behaviours

FR-060. When the noon release flag for a sitting is set, individual candidate checks (FR-010), school bulk requests (FR-030), and verification requests (FR-040) for that sitting begin returning grades; before the flag is set they return the not-yet-released response.

FR-061. When a result has been released, the system retains the ability to return that result (including its full amendment history) for a minimum of fifty years from the date of release.

FR-062. When any of the operations in sections 1–6 is performed, the system does not expose one candidate's data to another candidate, to an unauthenticated caller, or to a school or verifier that is not authorised for that candidate.

FR-063. When a PIN check, verification, or school bulk request is made under load up to the designed peak, the system continues to apply the same functional rules defined in FR-010 through FR-059; correctness is not relaxed under load.

FR-064. When a test harness replays a recorded sequence of PIN purchases, checks, verifications and amendments, the observable responses match the requirements above for every step in the sequence.

FR-065. When a PIN is exhausted (remaining count = 0) and a later amendment changes the candidate's grades, a subsequent check with that exhausted PIN still returns the exhausted-PIN message and does not return grades.

FR-066. When the system returns any refusal message defined in this document, the response body contains the refusal reason and does not contain subject grades or other candidate result data.

FR-067. When a candidate check, school bulk request, or verification request is received after the noon release flag has been cleared for a sitting (for example after the sitting is archived), the system continues to return the released grades for that sitting under the same rules as FR-010, FR-030 and FR-040.

FR-068. When an amendment officer views the audit trail for a candidate subject, the system displays the original release grade and every amendment in chronological order with officer identity, timestamp and evidence reference, matching the data returned by FR-056.

FR-069. When a PIN purchase, candidate check, school bulk, verification or amendment operation fails for a stated reason in this document, the system returns a stable, machine-readable reason code in addition to the human-readable message so that automated tests can assert the exact failure mode.

FR-070. When the same PIN is presented on web, USSD and SMS in succession while remaining count is greater than zero, each successful presentation decrements the count by one and returns the current grades, until the count reaches zero and subsequent presentations receive the exhausted-PIN response.

3.2 PRD (Product Requirement Document)

Product Requirements Document

Results Check, Certificate Verification & School Bulk Access

Audience: engineering, product, and commercial teams who build and sell the service

1. Users, the Job They Are Finishing, and a Good Outcome

Four primary user types. Roles that only read management reports are outside this document. Each row states the job in the user's language and the concrete outcome that counts as finished.

User	Job to finish	Good outcome (what they leave with)
Candidate	Buy a PIN and learn their nine subject grades on whatever phone and network they have	Within seconds of a valid PIN they see their own grades. Up to five checks. No second charge on their statement. Works on USSD, SMS or browser.
School (principal / exams officer)	Obtain the complete result list for every candidate under that school's centre number on results day	Authenticated download or portal view containing only that school's candidates, available from noon, matching what candidates see individually.
Verifier (employer / university)	Confirm a presented certificate matches a genuine released result for the named person and subjects	Authoritative yes/no plus the exact grades released (or current amended grades). Never shown other subjects, address, phone or centre.
Amendment officer (Council ops)	Correct a grade after remark or appeal, leave a permanent record of who acted and on what evidence, make the new grade the one future checks return	New grade live for every subsequent check and verification within minutes. Full audit trail queryable. Prior printed certificates are not rewritten.

Secondary commercial user — licensed PIN agents and bank channels. Job: sell a PIN without friction or chargeback risk.

Good outcome: payment succeeds once, PIN is issued once, agent is not held responsible for double charges.

Edge conditions each user must still finish the job under

- **Candidate on 2G / feature phone:** USSD menu and SMS reply must complete in fewer than four network round-trips; no dependency on JavaScript or high bandwidth.
- **Candidate whose payment was interrupted:** Retrying the same purchase must not create a second charge; the product layer supplies the idempotency the payment partner lacks.
- **School at noon:** Bulk request must not compete with the individual-check surge path; a separate, rate-controlled channel is required so one cannot starve the other.
- **Verifier requesting one subject:** Response contains only that subject's grade and validity; other subjects and personal contact data are never returned.
- **Amendment after a certificate was already verified:** Future checks and verifications return the new grade; the product does not attempt to recall or rewrite records already held by third parties.

2. Scope of the First Release (May Sitting)

In scope — must ship

- PIN purchase with idempotent payment handling (no double charge) and five-check entitlement.
- Result check over USSD, SMS and web returning the candidate's nine subject grades.
- Capacity path that absorbs the observed results-day peak ($\geq 2,500$ checks/sec sustained one minute; ≥ 1.4 m checks in the first hour) without raising monthly infrastructure cost.
- Certificate verification API and portal; returns only requested subject grades and a validity flag; data minimisation enforced at the response boundary.
- Amendment workflow requiring officer identity and evidence reference; new grade takes effect on all future checks and verifications within minutes.
- School bulk result list for the May sitting via authenticated download or portal, available from noon, on a channel separate from the individual surge path.
- All newly released results written so they remain verifiable for the fifty-year regulatory horizon; existing archive remains queryable.

Explicitly out of scope for first release

- Rewriting or invalidating certificates already printed or already verified when an amendment occurs. Future checks show the amendment; stored third-party history is not altered.
- Real-time bulk download that shares the hot path with individual candidate checks at noon.
- Self-service remark or appeal submission by candidates. That remains an offline Council process that feeds the amendment workflow.

- Native mobile apps. Supported surfaces are USSD, SMS and responsive web only.
- Full historical bulk export tools for researchers or open-data portals.
- Replacement of the payment partner. The existing partner stays; the product adds the idempotency and reconciliation layer on top of it.

Definition of done for each primary job (first release)

These are the concrete checks product and engineering use before calling a job “finished” for R1. They are not the release metrics in section 4; they are the per-feature acceptance gates.

- **Candidate job:** A test PIN purchased on a feature-phone USSD session returns all nine grades in under 8 seconds end-to-end; a forced payment retry with the same client reference produces exactly one charge and one PIN.
- **School job:** An authenticated school account can download a complete centre file within 15 minutes of the noon release flag; the file matches a sample of individual checks for that centre.
- **Verifier job:** A verification request for one named subject returns only that subject’s grade and a validity flag; a request that omits subject scope is rejected; sampled responses contain zero forbidden fields.
- **Amendment job:** An officer-submitted amendment appears on a subsequent candidate check and on a subsequent verification call within 5 minutes; the audit record contains officer identity, timestamp and evidence reference and cannot be deleted.

3. Release Sequence and Why This Order

Three releases. Sequencing is driven by irreversible calendar risk (May sitting), revenue protection (PIN integrity), and contractual growth (verification volume), not by feature attractiveness or engineering preference.

Release	Target window	What ships and the reason it is ordered here
R1 – Core	Live before May results day	PIN sale + idempotent payment, result check (USSD/SMS/web), results-day capacity path, amendment write path, basic verification endpoint. Reason: without these the May sitting cannot be released on the new system and double-charge leakage continues. Every other feature is subordinate to this hard date.
R2 – Scale & schools	Within 6 weeks after May	Hardened school bulk channel, verification portal UI, rate limits and monitoring sized for the rise toward 500 k verifications/month, operational run-books. Reason: schools and verifiers become the loudest users immediately after first release; capacity and support must be proven before employer traffic ramps.
R3 – Harden & extend	Before November sitting	November sitting support, refined audit reporting for the regulator, performance tuning from May telemetry, deferred non-critical UX. Reason: second sitting is smaller but must not regress; regulator evidence and operational maturity are the remaining hard constraints before the year closes.

No parallel “nice-to-have” track. Anything not listed in R1–R3 stays on the backlog until after November. With six engineers, concurrent streams increase delivery risk rather than speed.

Dependency rule: R2 may not start hardening work that assumes R1 capacity numbers until those numbers have been measured on the live May release. R3 may not change the verification contract until R2 has demonstrated sustained 50 k verifications/day.

4. Metrics That Decide Whether a Release Worked

Each release is judged by a short list of numbers measured after the fact. Missing the numbers means the release is treated as failed for commercial and operational purposes even if the code shipped on time.

R1 success metrics (measured in the week after May results day)

1. **Capacity:** ≥ 1.4 million successful result checks in the first hour; at least one continuous minute at $\geq 2,500$ checks/sec with error rate $< 0.5\%$.
2. **Payment integrity:** Zero double charges attributable to the product layer (reconciled against the payment-partner settlement file).
3. **Channel reach:** $\geq 15\%$ of all checks arrive via USSD or SMS, proving the feature-phone path is live and used.
4. **Cost ceiling:** Infrastructure spend for the results calendar month $\leq \text{N}14$ million.
5. **Amendment correctness:** 100 % of amendments applied in the first month appear on subsequent checks within 5 minutes; every amendment carries a complete audit record.

R2 success metrics (measured six weeks after May)

6. **School delivery:** $\geq 90\%$ of the 21,000 schools that request a bulk list receive a complete, matching file within 2 hours of the noon release.
7. **Verification readiness:** Sustained 50,000 verification requests per day with p95 latency < 2 s and zero data-minimisation violations in sampled audit reviews.
8. **Support load:** Fewer than 200 open tickets attributable to product defects 48 hours after results day.

R3 success metrics (measured after November sitting)

9. **Second-sitting parity:** Same capacity, payment-integrity and cost metrics as R1, applied to the smaller November volume.
10. **Regulatory evidence:** Regulator can retrieve any released result from the May or November sitting together with its full amendment history in a single authenticated query.

11. Verification growth path: Demonstrated ability to run at 200,000 verifications/month without infrastructure cost increase, giving confidence for the contracted 500,000/month trajectory.

Commercial rule: Sales and account teams may quote only metrics that have already been measured and passed. Pre-release claims are limited to the in-scope list in section 2. No metric from a later release may be promised while an earlier release's metrics remain unmet.

3.3 BRD (Business Requirement Document)

Business Requirements Document

Replacement of the Examination Results and Certificate Verification System

Prepared for the Board of the National Secondary Certificate Council

1. The Problem in Business Terms

The Council currently runs its results release, certificate printing support, and third-party verification services on a system built in 2009. That system was never designed for the volume of demand that now occurs on results day, nor for the growth in employer and university verification requests that the Council has already contracted to deliver. It also has no protection against double-charging candidates when they buy result-checker PINs. The organisation therefore faces three simultaneous pressures: operational risk on the single most public day of the year, financial leakage from payment failures, and an inability to honour a signed growth agreement with the national employers' association without increasing infrastructure spend beyond the fixed monthly ceiling.

2. Who Suffers Today and What It Costs

Candidates. Last year 9,100 candidates were charged twice for a single PIN purchase because the payment partner does not support an idempotency key. Each double charge is ₦3,500. The direct cash cost to candidates was approximately ₦31.9 million. Beyond the money, the experience damages trust in the Council on the day results are most visible.

Schools and candidates on results day. In the first hour after release last year, 1.4 million result checks arrived. The busiest ten minutes averaged 930 checks per second and the peak minute reached roughly 2,900. The present system cannot absorb that load without queues or outages. Schools (21,000 of them) also expect their full candidate lists at noon; any delay or failure is felt immediately by principals and parents.

Employers, universities and the Council's reputation. Verification requests stand at 40,000 per month today. A signed agreement is expected to lift this to 500,000 per month within a year. The existing platform has no proven capacity path for that growth inside the fixed budget. Failure to deliver would breach a commercial commitment and expose the Council to reputational and contractual risk.

The Council's own finances and risk position. Infrastructure is capped at ₦14 million per month for the next two years. Any design that scales cost with results-day traffic will be refused. Continuing with the current system therefore preserves a known failure mode on results day while leaving the double-charge leakage and the verification growth obligation unaddressed.

3. What Success Looks Like (Measurable)

Success will be judged against one primary number that can be checked after the fact:

On the next results-day release, the Council will serve at least 1.4 million result checks in the first hour, including a sustained peak of at least 2,500 checks per second for one continuous minute, with zero payment double-charges and with infrastructure cost remaining at or below ₦14 million for that calendar month.

Secondary checks that will also be reported: (a) zero verified certificates that later prove to have been amended without a full audit trail; (b) every released result still reproducible and verifiable five years after go-live (the start of the fifty-year regulatory obligation); and (c) the system remaining usable over USSD/SMS on 2G networks so that candidates without smartphones are not excluded.

4. Options Considered

1. **Do nothing.** Keep the 2009 system. Cost: continues the ₦31.9 million annual double-charge leakage (and rising), accepts results-day queues or outages as normal, and leaves the Council unable to meet the contracted verification growth. Risk of public failure on results day remains high. This option requires no new spend but transfers all operational and reputational risk onto the Board.
2. **Scale the current architecture with cloud auto-scaling.** Buy enough elastic capacity to absorb a 100–150× surge. This would satisfy the Registrar's "no queue" demand but would make monthly infrastructure cost variable and almost certainly exceed the ₦14 million fixed ceiling on results month. Rejected because it violates the Finance Director's non-negotiable constraint.
3. **Build a new system sized only for average load and throttle results day.** Cheapest to run, but directly contradicts the Registrar's requirement that every candidate sees their result the moment it is released. Also leaves schools unable to obtain bulk lists at noon. Rejected as operationally and politically unacceptable.
4. **Replace the system with a design that separates the results-day surge from the year-round transactional work.** Pre-generate and cache results so that the peak can be served from static or near-static infrastructure whose cost does not rise with traffic; keep a separate, slower, fully audited path for verification, amendments and PIN sales. This is the only option that simultaneously respects the fixed budget, eliminates double charges, supports USSD/SMS, and gives a realistic path to 500,000 verifications per month.

5. Recommendation

The Board is recommended to authorise the design and build of a replacement system under option 4 above. The architecture will deliberately accept that not every stakeholder demand can be met in full: instantaneous universal visibility of amendments on already-printed or already-verified certificates is impossible without rewriting history, and schools will receive bulk lists via a controlled, non-surge path rather than competing with individual candidates at the exact moment of peak load. These trade-offs are explicit and are judged preferable to either uncontrolled cost growth or continued public failure risk.

6. What the Council Is Committing to Spend and to Risk

Spend. Infrastructure remains capped at the existing ~~£~~14 million per month for the first two years; the design will be required to stay inside that envelope even on results day. Build cost is to be met from the already-approved project budget for a six-engineer team over nine months to the May sitting. No additional recurrent operational expenditure is requested beyond the current ceiling.

Risk the Council accepts. (1) Delivery risk: a nine-month timetable with a small team means scope must be tightly controlled; features that are not essential to the measurable success criterion above will be deferred. (2) Amendment visibility: once a certificate has been verified or printed, a later amendment will be fully audited and available on subsequent checks, but will not automatically invalidate or rewrite every prior verification record already held by third parties. (3) Schools bulk access: full candidate lists will be available on results day, but via a dedicated channel that does not share the individual-check surge path, so that one does not starve the other.

Risk the Council transfers away. The chronic double-charge leakage ends. Results-day capacity is designed for the observed peak rather than hoped for. The verification growth obligation becomes deliverable inside the fixed budget. Candidates on 2G feature phones remain first-class users.

Decision required of the Board: *Approve the replacement programme under the fixed ~~£~~14 million monthly infrastructure ceiling, accept the explicit trade-offs on amendment propagation and school bulk delivery, and hold the project to the single measurable success criterion stated in section 3.*

3.4 WBS (Work Breakdown Structure)

Work Breakdown Structure

Nine months to May sitting | Team of six | First-release scope only

Audience: delivery lead, engineers, and anyone who must decide what fits. Every package ≤ 2 engineer-weeks.

0. Capacity and honesty check

Calendar: 9 months ≈ 39 working weeks. Team: 6 engineers. Gross capacity ≈ 234 engineer-weeks. After 15 % loss for meetings, onboarding, leave and integration friction the usable capacity is treated as 200 engineer-weeks. The packages below total 168 engineer-weeks of estimated effort. That leaves ~ 32 engineer-weeks of buffer for defect fix, load testing and release hardening. The plan therefore fits a team of six before the May sitting. Nothing from the first-release scope in the PRD/FRD is cut; items already marked out-of-scope in the PRD remain out.

Columns: ID · Package · Deliverable · Depends on · Engineer type · Est. (eng-wk). Engineer types: BE = backend, FE = web front-end, INT = integration (USSD/SMS/payment), INF = infrastructure/DevOps, QA = test/automation, FS = full-stack.

1. Foundations (Weeks 1–6)

ID	Package	Deliverable	Depends on	Type	Est.
W1.1	Repo, CI, env parity	Git repo, CI pipeline, staging + prod skeleton, secrets management	—	INF	1.5
W1.2	Identity & auth service	Service accounts for schools, verifiers, officers; token issue/validate	W1.1	BE	2.0
W1.3	Core data model & migrations	Candidates, subjects, results, PINs, amendments schema + migration scripts	W1.1	BE	2.0
W1.4	Audit log store	Append-only audit table/API for checks, verifications, amendments	W1.3	BE	1.0
W1.5	Observability baseline	Metrics, structured logs, error tracking, basic dashboards	W1.1	INF	1.5
W1.6	Test data & fixtures	Synthetic candidates, PINs, centres for local and staging tests	W1.3	QA	1.0

2. PIN Purchase & Payment Integrity (Weeks 4–12)

ID	Package	Deliverable	Depends on	Type	Est.
W2.1	Payment partner adapter	Client that calls partner charge API; records client reference	W1.1	INT	2.0
W2.2	Idempotent purchase API	FR-001–FR-004: unique client ref $\rightarrow$ one charge, one PIN, remaining=5	W2.1, W1.3	BE	2.0
W2.3	PIN store & remaining count	Atomic decrement of remaining checks; concurrent last-check safety (FR-015)	W1.3	BE	1.5
W2.4	Reconciliation job	Daily match of settlement file to PINs (FR-005); alert on mismatch	W2.2	BE	1.5
W2.5	PIN status lookup	FR-008: status by client reference without new charge	W2.2	BE	0.5
W2.6	Payment & PIN tests	Automated tests for double-submit, decline, in-progress, reconciliation	W2.2–W2.4	QA	2.0

3. Result Check Core (Weeks 6–16)

ID	Package	Deliverable	Depends on	Type	Est.
W3.1	Results ingest pipeline	Load released results for a sitting; set noon release flag	W1.3	BE	2.0
W3.2	Check API (web path)	FR-010–FR-019: validate PIN, return grades, decrement, refusals	W2.3, W3.1	BE	2.0
W3.3	Web check UI	Browser form: enter PIN, display grades or refusal message	W3.2	FE	1.5
W3.4	Check audit writer	FR-016: record PIN, channel, timestamp, remaining after check	W1.4, W3.2	BE	0.5

W3.5	Check API tests	Exhausted, invalid, not-released, concurrent last-check, data minimisation	W3.2	QA	2.0
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4. USSD & SMS Channel (Weeks 10–18)

ID	Package	Deliverable	Depends on	Type	Est.
W4.1	USSD gateway adapter	Session handler; prompt for PIN; ≤4 round-trips (FR-020, FR-025)	W3.2	INT	2.0
W4.2	SMS gateway adapter	Keyword + PIN parse; single SMS reply with grades or refusal (FR-023)	W3.2	INT	1.5
W4.3	Channel audit & timeouts	FR-024, FR-027, FR-028: channel tag, timeout no-op, malformed PIN	W4.1, W4.2	INT	1.0
W4.4	USSD/SMS tests	End-to-end scripts on test short-code and USSD simulator	W4.1–W4.3	QA	1.5

5. Results-Day Capacity Path (Weeks 12–22)

ID	Package	Deliverable	Depends on	Type	Est.
W5.1	Pre-generated result cache	Build static/near-static result payloads at release; serve without hot DB	W3.1	BE	2.0
W5.2	CDN / edge serving	Edge cache for check responses; invalidation on amendment	W5.1	INF	2.0
W5.3	PIN gate in front of cache	Validate PIN + remaining count before serving cached grades	W2.3, W5.1	BE	1.5
W5.4	Load-test harness	Replay peak profile (≥2,500/sec minute; 1.4 m first hour)	W5.2, W5.3	QA + INF	2.0
W5.5	Cost & capacity runbook	Confirm monthly infra ≤ ₦14 m under peak; scale limits documented	W5.4	INF	1.0

6. Verification API (Weeks 14–24)

ID	Package	Deliverable	Depends on	Type	Est.
W6.1	Verification API	FR-040–FR-048: auth, subject-scoped grades, data minimisation, audit	W1.2, W3.1	BE	2.0
W6.2	Verifier portal (basic)	Login + request form + result display for employers/universities	W6.1	FE	1.5
W6.3	Verification tests	Minimisation samples, auth failure, mixed subjects, concurrent reads	W6.1	QA	1.5

7. Amendment Workflow (Weeks 16–26)

ID	Package	Deliverable	Depends on	Type	Est.
W7.1	Amendment write API	FR-050–FR-059: officer + evidence, immutable audit, 5-min propagation	W1.2, W1.4, W3.1	BE	2.0
W7.2	Cache invalidation on amend	Invalidate edge/cache entries so FR-051 holds for checks & verification	W5.2, W7.1	BE	1.0
W7.3	Officer UI + history query	Submit amendment; regulator history view (FR-056, FR-068)	W7.1	FE	1.5
W7.4	Amendment tests	Concurrent amend, no-evidence refusal, propagation timing, bulk consistency	W7.1–W7.3	QA	1.5

8. School Bulk Download (Weeks 18–28)

ID	Package	Deliverable	Depends on	Type	Est.
W8.1	Bulk export service	FR-030–FR-038: centre-scoped file, post-release only, separate channel	W1.2, W3.1	BE	2.0
W8.2	School portal download	Auth school login + download button; rate-controlled path	W8.1	FE	1.0
W8.3	Bulk consistency tests	Match individual checks; amendments reflected; wrong-centre refusal	W8.1	QA	1.0

9. Hardening, Archive & Release (Weeks 24–36)

ID	Package	Deliverable	Depends on	Type	Est.
W9.1	Archive retention path	FR-061: write path that keeps results + amendment history ≥50 years	W1.3, W7.1	BE	1.5
W9.2	End-to-end release rehearsal	Full dress rehearsal: ingest → noon flag → peak check → amend → bulk	W5.4, W8.1	QA + FS	2.0
W9.3	Security & access review	Authz matrix, no cross-candidate leakage (FR-062), secrets audit	W1.2, all APIs	BE + INF	1.5
W9.4	Runbooks & on-call	Incident runbooks for results day, payment mismatch, USSD outage	W5.5	INF	1.0
W9.5	Production cut-over	DNS, feature flags, final load check, go-live for May sitting	W9.2, W9.3	INF + FS	2.0

10. Totals and cut decision

Sum of package estimates: 168 engineer-weeks. Usable capacity ($6 \times 39 \text{ weeks} \times 0.85$): ~200 engineer-weeks. Buffer remaining: ~32 engineer-weeks for defect fix, peak-day support and contingency.

Verdict: The first-release scope fits a team of six before the May sitting. No in-scope package is cut. Items already listed as out of scope in the PRD (native apps, self-service appeals, payment-partner replacement, rewriting already-verified certificates, shared hot-path bulk) stay out and are not estimated here.

If buffer is consumed early: Defer W6.2 (verifier portal UI) and serve verification via API only for May; defer W8.2 (school portal polish) and deliver bulk as authenticated file URL only. Core check, PIN integrity, capacity, USSD/SMS, amendment write and school bulk file remain non-negotiable for the May sitting.

4. Part B: Architecture and the Decisions

4.1 Ranked quality attributes

Each attribute is a **scenario with a number**. Lower ranks are what we agreed to be worse at.

Rank	Attribute	Scenario (measurable)	Sacrifice forced by this rank
1	Results-day availability inside cost ceiling	≥1.4M checks in first hour, ≥2,500/s one minute, error rate <0.5%, month spend ≤ ₦14,000,000	No cost-scaling with traffic; residual Registrar queue risk if pre-gen fails
2	Payment integrity	Zero double charges attributable to our layer	We own partner timeout and reconciliation
3	Integrity & reproducibility	Result reconstructable as of any past date for 50 years; amendments name officer + evidence	No in-place result updates; no rewrite of past external verifications
4	Privacy (DPO)	Verification returns only requested subjects + validity	No full-candidate dump for verifiers
5	Channel reach (2G)	USSD/SMS live; target ≥15% of checks via USSD/SMS on results day	Extra USSD/SMS edge
6	Amendment propagation (future reads)	New grade on subsequent checks/verifications within 5 minutes	Not “everywhere including already-verified certificates”
7	School bulk completeness	≥90% of requesting schools get complete file within 2 hours of noon	Bulk not on candidate surge path

By ranking (1)–(3) above uniform simplicity and “amendment rewrites history,” we accept being worse at single-path operations and at the Head of Operations’ literal demand.

4.2 Architectural pattern chosen and 2 rejected

Plain language

The system has two very different jobs:

For eleven months: quiet transactional work (PIN sales, occasional checks, verifications, amendments) at under 20 requests per second.

For roughly one hour once a year: a spike of about 930 checks per second on average in the first ten minutes, peaking near 2,900 per second, while the monthly infrastructure bill must stay at or below ₦14 million.

A single pattern that treats every request the same way will either blow the budget on results day or force queues on the day the Registrar cares about most. So the platform is split:

Chosen pattern: Hybrid read-optimised surge path + transactional core

Surge path (individual result checks on results day): Results for a sitting are pre-generated (or materialised) before noon and served from cache/edge. The expensive work is done once, ahead of time. Serving a check is then mostly a PIN validation + a cache lookup. Cost stays largely flat even at 2,900/s.

Transactional core (everything else): Ordinary services for PIN purchase, verification, amendment, school bulk, and steady-state checks. These stay simple and do not try to absorb the peak.

Pattern	Why it was considered	Specific property of <i>this</i> system that ruled it out
1. Classic 3-tier (or modular monolith) + cloud auto-scaling on results day	Familiar, easy to staff with six people, scales “automatically” when traffic rises.	Finance Director’s rule is absolute: cost must not rise with results-day traffic. Auto-scaling that multiplies instances with the 145× surge (2,900/s vs <20/s) makes the May invoice exceed ₦14 M. A rejection reason that only said “auto-scaling is expensive” would be generic; the deciding property is the fixed monthly ceiling against a known 100–150× spike concentrated in one hour.

2. Full event-sourcing / CQRS everywhere (every write is an event, every read is a projection)

Excellent for 50-year audit and “as of date” reproducibility; amendments become natural.

Most of the year the load is <20/s and amendments are only ~475 per sitting. Paying the full event-sourced complexity (projections, eventual consistency, operational tooling) for the quiet 11 months, with only six engineers and a hard May deadline, is disproportionate. The deciding property is low steady-state load + small amendment volume + fixed team size, not a generic dislike of event sourcing.

What the ranking costs

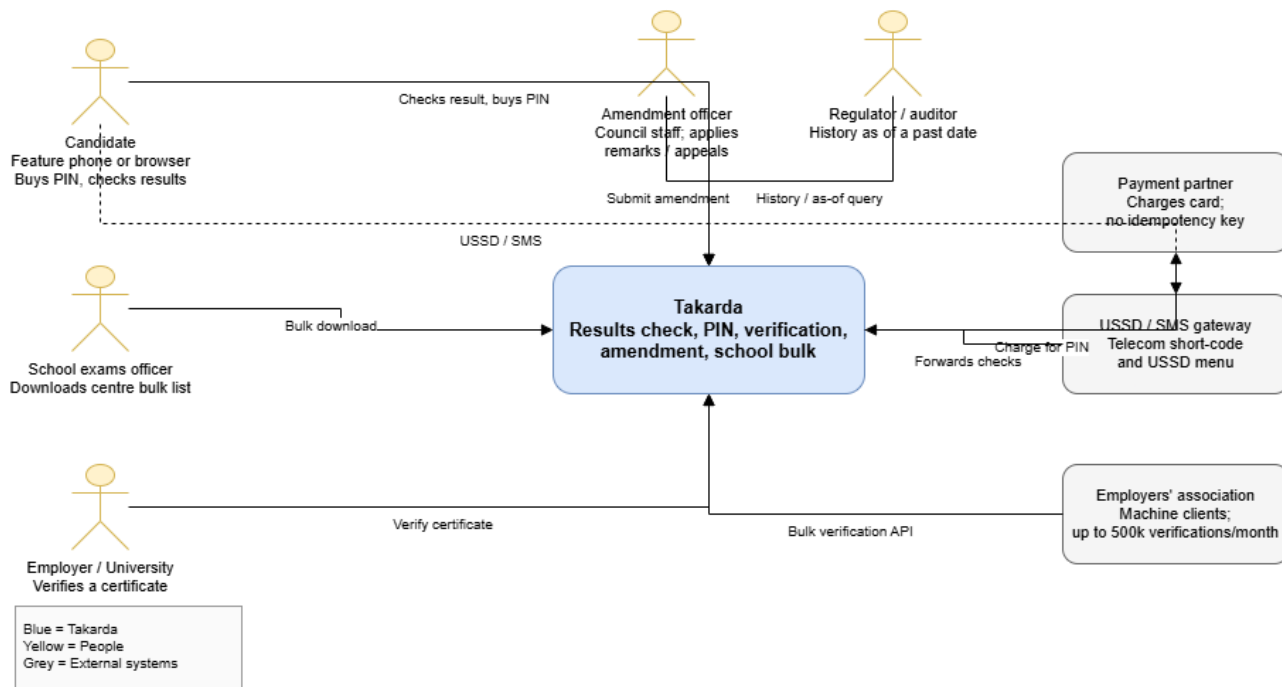
We are deliberately worse at: (a) “one uniform mental model for every request”, and (b) “instant global visibility of an amendment on every already-issued third-party verification”. We are better at: staying inside ~~N~~14 M on results month, giving candidates a no-queue path at the observed peak, and keeping a clear audit trail for the Regulator.

4.3 C4 diagrams

Diagrams are given as Mermaid so you can put the source files in the repository (not screenshots). Each is followed by what it deliberately leaves out.

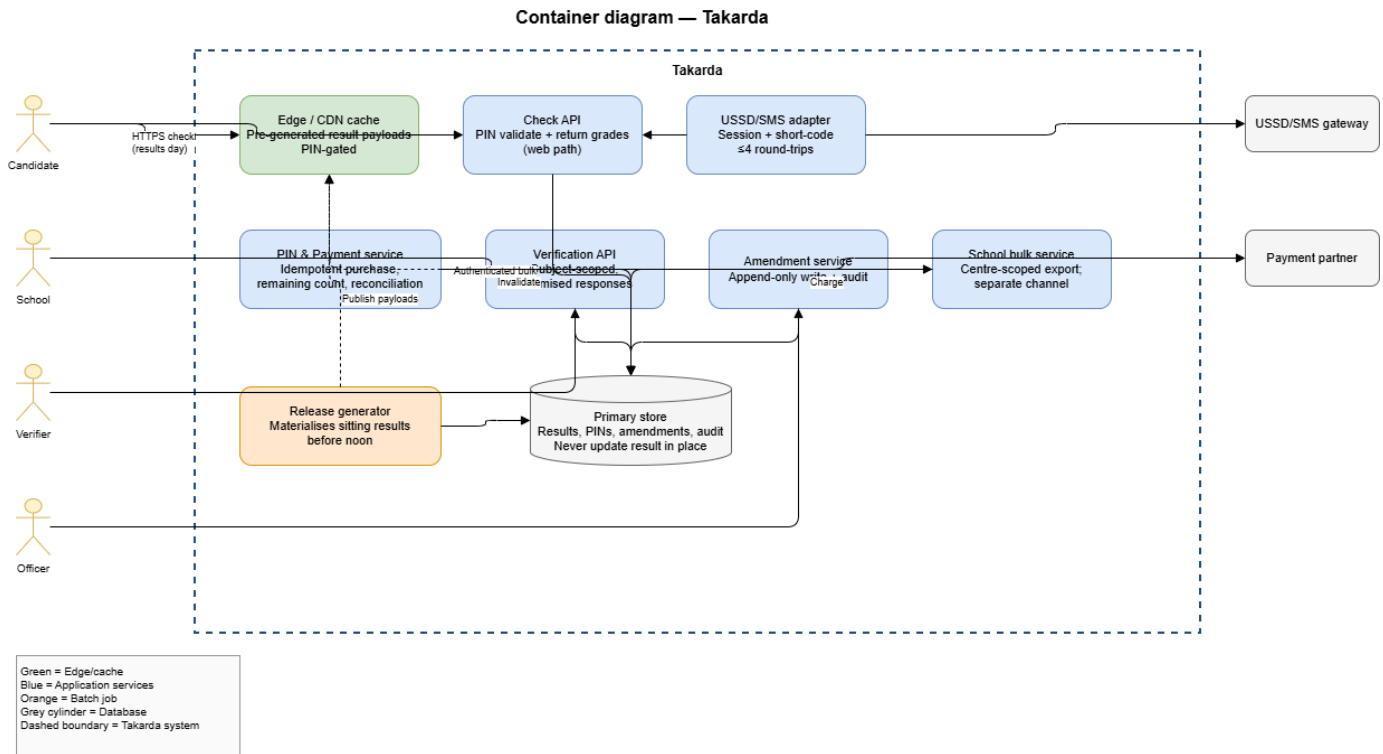
System Context

System Context — Takarda (National Secondary Certificate Council)



Deliberately left out: Internal Council reporting, certificate printing systems, bank agent UIs beyond “they sell a PIN”, and any identity provider the Council already owns. Those are real but not the design surface we are deciding.

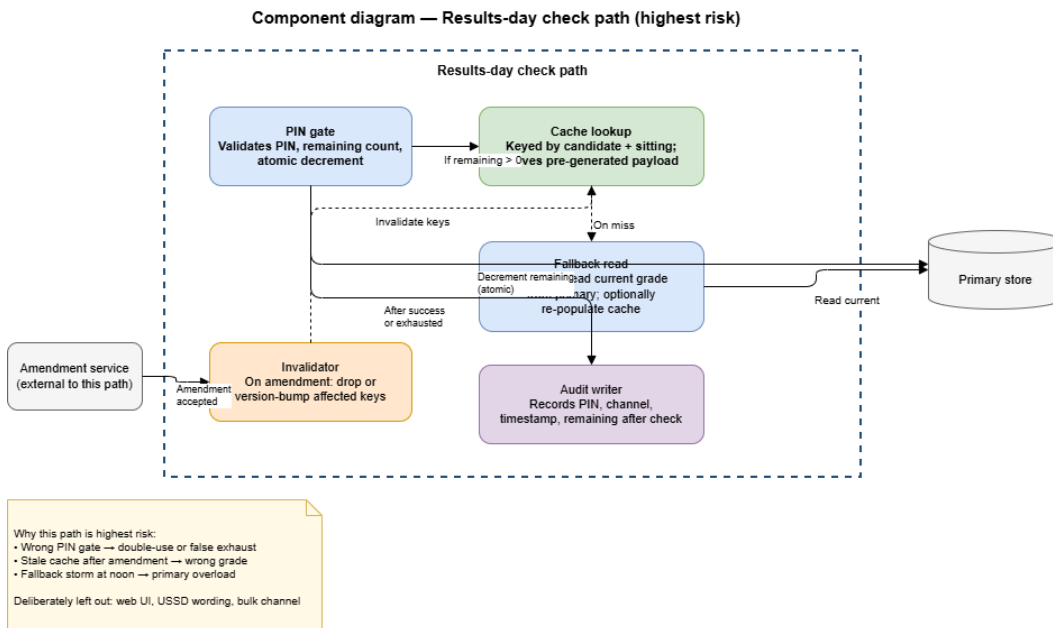
Container Diagram



Deliberately left out: Exact number of replicas, CI/CD, monitoring stack, and the internal admin UI chrome. Those matter operationally but are not the architectural fork.

Component diagram — the part most likely to be got wrong

The riskiest part is the **results-day individual check path**: PIN gate + cache + invalidation on amendment. Getting this wrong either creates queues or double-serves stale grades after an amendment.



Deliberately left out: Web UI rendering, USSD menu wording, and the bulk channel (it is intentionally a different path so it cannot starve this one).

4.4 Architecture Decision Records (minimum 8)

Format for each: Context → Decision → Alternatives → Consequences → **Signal that would reverse this.**

At least two are flagged as **one-way doors**.

ADR-01 — Split surge path from transactional core

One-way door: yes (once results are pre-generated and clients depend on cache behaviour, switching back to pure elastic compute is a redesign, not a config change).

- **Context:** 145x spike for ~1 hour vs <20/s for 11 months; ₦14 M/month fixed.
- **Decision:** Individual result checks on results day are served from a pre-generated / edge-cached path. All other work uses ordinary transactional services.
- **Alternatives:** (a) Auto-scale one path for everything; (b) Stagger release by state/surname to flatten the peak.
- **Consequences:** Better — cost stays inside ceiling; candidates get no-queue path at observed peak. Worse — two paths to build, test and invalidate; generation must finish before noon.
- **Signal that would reverse this:** Observation: monthly infrastructure invoice for a results month exceeds ₦14 M or measured p99 check latency on the cache path exceeds 3 s while error rate stays <0.5%. Watched by: Finance Director (invoice) + Head of Operations (latency dashboard).

ADR-02 — Result rows are never updated in place; amendments are append-only

One-way door: yes (once external verifiers and the regulator rely on history, introducing in-place updates destroys reproducibility).

- **Context:** Regulator requires “as it stood on any past date” + named officer; Ops wants “new grade everywhere immediately”.
- **Decision:** A released subject-result is immutable. An amendment is a new record (previous grade, new grade, officer, evidence, timestamp). “Current” grade is the latest amendment or the original if none.
- **Alternatives:** (a) Update the result row in place and keep a side log; (b) Soft-delete + insert.
- **Consequences:** Better — true reproducibility and audit. Worse — queries for “current” must resolve the latest amendment; storage grows with amendments (~475/sitting, small).
- **Signal that would reverse this:** Observation: regulator rejects an as-of-date query as non-reproducible, or amendment volume exceeds 50,000 per sitting (two orders of magnitude above brief). Watched by: Regulator liaison + engineering lead.

ADR-03 — School bulk on a dedicated channel

- **Context:** 21,000 schools want lists at noon; candidate peak is 2,900/s on the same calendar second.
- **Decision:** Bulk export is a separate service and rate-controlled path. It does not share the edge cache or the individual-check hot path.
- **Alternatives:** (a) Same API and path as individual checks; (b) Pre-email all 21,000 files overnight.
- **Consequences:** Better — candidate surge is protected. Worse — schools may wait minutes; bulk has its own capacity plan.
- **Signal that would reverse this:** Observation: >10% of schools fail to receive a complete file within 2 hours of noon on results day (PRD R2 metric). Watched by: Head of Schools + support lead.

ADR-04 — Idempotency and reconciliation live on our side of the payment boundary

- **Context:** Partner accepts no idempotency key; 9,100 double charges last year.
- **Decision:** We generate and store a client reference; we own the state machine for in-progress / success / failure; we reconcile against the settlement file and drive refunds without a human spreadsheet.
- **Alternatives:** (a) Wait for the partner to add idempotency; (b) Accept double charges and handle manually.
- **Consequences:** Better — zero double charges attributable to us (BRD success metric). Worse — we carry timeout and “unknown outcome” complexity.
- **Signal that would reverse this:** Observation: settlement reconciliation finds >0.1% unmatched successful charges for two consecutive sittings. Watched by: Finance + PIN service owner.

ADR-05 — Protocol and authentication split by consumer type

- **Context:** Candidate on 2G (round-trip cost), school bulk (large payload), employers' association (machine, 500k/month).
- **Decision:**
 - Candidates: USSD + SMS (and HTTPS for those who have it).
 - Schools & human verifiers: HTTPS + token.
 - Employers' association: HTTPS + mTLS (machine identity).
- **Alternatives:** (a) One HTTPS API for everyone; (b) USSD for all.
- **Consequences:** Better — 2G users are not punished by TCP+TLS handshake cost; machines get stronger identity. Worse — three edges to operate.

- **Signal that would reverse this:** Observation: USSD/SMS share of checks falls below 5% for two sittings **and** support tickets show candidates blocked only by channel, not by coverage. Watched by: product + channel metrics.

ADR-06 — Verification responses are field-minimised at the boundary

- **Context:** DPO forbids other subjects, address, phone, centre; volume heading to 500k/month.
- **Decision:** Verification API accepts explicit subject list (or certificate scope) and returns only those grades + validity. No “full candidate” endpoint for verifiers.
- **Alternatives:** (a) Return full candidate record and let client filter; (b) Separate “rich” API for trusted partners.
- **Consequences:** Better — DPO compliance by construction. Worse — verifiers who wanted a dump must call more precisely.
- **Signal that would reverse this:** Observation: audit sample finds any response containing a forbidden field. Watched by: DPO + verification service owner.

ADR-07 — Primary database is a single logical system of record with read replicas; no sharding for v1

- **Context:** 216M rows today, 50-year retention, six engineers, May deadline.
- **Decision:** One primary for writes; read replicas for check/verify/bulk reads that can tolerate replica lag. Partitioning by sitting (or sitting + centre) is allowed later; sharding across machines is deferred.
- **Alternatives:** (a) Shard by candidate ID from day one; (b) Separate database per sitting.
- **Consequences:** Better — operational simplicity for six people; cross-sitting regulator queries stay easy. Worse — a single primary is a scaling ceiling we must watch.
- **Signal that would reverse this:** Observation: primary CPU or I/O saturates under steady-state verification load (500k/month) **or** release load of 1.9M results exceeds the pre-noon window. Watched by: infrastructure owner.

ADR-08 — First release does not include native apps, self-service appeals, or payment-partner replacement

- **Context:** CE constraint — May, six engineers.
- **Decision:** Explicit out-of-scope list (already in PRD). Native apps, candidate-initiated appeals, and replacing the payment partner wait until after November at earliest.
- **Alternatives:** (a) Thin native wrappers; (b) Build partner replacement in parallel.
- **Consequences:** Better — May is achievable. Worse — some stakeholders wait.
- **Signal that would reverse this:** Observation: WBS buffer is exhausted before R1 hardening **or** CE moves the May date. Watched by: delivery lead.

4.5 Trade-off register

One row per ADR. Every row has a **named victim**.

ADR	Makes better	Makes worse	Named victim who feels the cost
01 Surge/transactional split	Cost stays ≤ ₦14 M at peak; candidates get no-queue path	Two paths to build and invalidate; residual queue if generation fails	Registrar (residual queue risk) and engineering (two paths)
02 Append-only results	50-year reproducibility; officer trail	“Current grade” resolution; Ops cannot rewrite past verifications	Head of Operations
03 Dedicated bulk channel	Candidate surge protected	Schools may wait minutes	Head of Schools
04 Our-side idempotency	Zero double charges	Timeout/unknown-outcome complexity owned by us	PIN service owner / engineering
05 Protocol split	2G users viable; strong machine auth	Multiple edges to operate	Channel/ops engineers (and candidates if USSD is down)
06 Minimised verification	DPO compliance by construction	No full-candidate dump for verifiers	Employers who wanted richer payloads
07 No sharding v1	Simple operations for six people	Single-primary ceiling	Infrastructure owner (must watch saturation)
08 Scope cut for May	May is achievable with six engineers	Deferred features	Stakeholders waiting for native apps / self-service appeals / partner replacement

4.6 What we are not building

Capability a reasonable person might expect	Why it is out	Revisit condition
Native mobile apps (iOS/Android)	Six engineers, nine months; USSD/SMS/web already cover 2G and smartphone users	After November sitting, if USSD/SMS share is healthy and buffer remains
Self-service remark/appeal by candidates	Appeals remain an offline Council process that feeds the amendment API; building intake + workflow would consume the May buffer	When amendment volume or political pressure makes manual intake the bottleneck
Replacement of the payment partner	Partner change is a commercial and integration project of its own; we fix double-charges on our side of the boundary	If reconciliation error rate stays above the ADR-04 threshold for two sittings or partner contract ends
Real-time invalidation of already-issued third-party verifications	Impossible as written without controlling every external copy; conflicts with reproducibility	Only if law or regulator later requires a formal recall channel (new requirement)
Full historical open-data export for researchers	Not required for May success metric; large export surface vs DPO	After regulator confirms export rules and R3 hardening is done

5. Part C – The Wire

5.1 Protocol choice per consumer

Plain language

Three consumers could not be more different:

1. A candidate on a feature phone on 2G — every round-trip costs time and money; TCP + TLS handshakes before any data moves are expensive.
2. A school pulling hundreds of rows once at noon — large payload, authenticated, but not 2,900 requests per second.
3. The employers' association — machines in a data centre, up to 500,000 calls per month, contractual, needs strong identity.

One protocol for all three would either punish the candidate or weaken the machine. We choose per consumer.

Decision table

Consumer	Protocol	Transport-layer decisions	Application-layer decisions	What breaks if wrong
Candidate (2G / feature phone)	USSD + SMS (HTTPS only as extra for those who have data)	Avoid new TCP+TLS per check. USSD is session-oriented on the operator network; SMS is one request / one reply. Target ≤ 4 network round-trips end-to-end (FR-025).	Same domain rules as web check (PIN, remaining, grades, refusals). Menu/keyword only.	If we forced HTTPS-only: many candidates on 2G cannot complete a check; handshake cost alone can exceed the time they will wait. Head-of-line blocking on a single TCP connection would also serialise their retries.
School bulk	HTTPS	Normal TLS. Keep-alive encouraged so a school that downloads once does not pay full handshake again for status. Bulk is not on the individual-check edge path.	Resource: centre + sitting. Auth: bearer token bound to centre(s). Rate-limited dedicated channel.	If bulk shared the surge path: 21,000 centres compete with 2,900 checks/s and either candidates or schools lose.
Employers' association (machine)	HTTPS	TLS required. mTLS (client certificate) preferred for machine identity. Connection reuse / HTTP keep-alive / HTTP/2 multiplexing to avoid head-of-line blocking under 500k/month.	Same minimised verification resource as human verifiers. Token optional as secondary; certificate is primary identity.	If we used only bearer tokens for machines: key distribution and rotation become operational risk at volume; a leaked token is harder to attribute than a client cert.

2G / handshake cost (why USSD/SMS wins for candidates)

On a cold HTTPS connection the client typically pays:

- TCP handshake (1 RTT)
- TLS handshake (1–2 RTTs)
- then the HTTP request (1 RTT)

On 2G, RTT can be hundreds of milliseconds to >1 s. Three RTTs before any grade appears is often worse than a USSD session that stays on the operator's signalling path. Keep-alive helps **repeat** checks on the same phone, but the first check still pays the full cost — and many candidates check once. USSD/SMS avoid that tax for the primary audience the Council refused to abandon.

Head-of-line blocking: On HTTP/1.1 one slow response blocks the next on the same connection. For the employers' association at volume we prefer HTTP/2 (or separate connections with limits) so one slow verification does not stall others. For USSD this is irrelevant; each session is short-lived.

5.2 Transport security

Plain language

TLS protects data in transit. Who presents a certificate, where TLS ends, and whether the client is a person or a machine are different decisions.

Decisions

Topic	Choice	Reason specific to this system
Where TLS terminates	At the edge / load balancer in front of Takarda services (for HTTPS consumers). USSD/SMS are terminated by the telecom gateway; we treat the gateway as a trusted path with mutual controls.	We do not terminate TLS on every application instance (operational cost for six engineers). Edge termination is enough for schools and verifiers.

Candidate (HTTPS)	Server certificate only (ordinary TLS). Candidate authenticates with PIN , not with a client certificate.	Candidates cannot manage client certs on feature phones.
School / human verifier / officer	Server TLS + bearer token (school/verifier/officer identity).	Humans; token issuance fits existing agent/school account processes.
Employers' association	Server TLS + mTLS client certificate as primary machine identity. Bearer optional as additional app-level auth.	Machine-to-machine on a contract; certificate identity matches "data centre talking to data centre".
Payment partner	Outbound TLS from our PIN service to partner; we present our client identity as required by partner.	We are the client on that leg.

Assumption: The Council's existing relationship with the USSD/SMS aggregator includes network-level authentication of the short-code path. If that is false, we must add an application-level HMAC or similar on the gateway→Takarda hop (ASSUMPTION: aggregator path is locked down by contract and IP allowlist).

5.3 The API contract (HTTP)

Plain language

Only HTTPS consumers use this API (schools, verifiers, officers, PIN agents, employers' association). Candidates on USSD/SMS hit the same **domain rules** through an adapter, not these URLs.

Rules used when designing:

- URLs name **resources** (nouns), not actions.
- Methods carry the verb; we state safe / idempotent / neither.
- Parameters are named, typed, required or not, and located (path = identity, query = filter, body = content).
- Every outcome including failure has a status code.
- Version is in the path (/v1). Old major versions stay live for a published period so a school on last year's client does not break overnight.

Resource map (summary)

Resource	Method	Safe?	Idempotent?	Role
/v1/pins	POST	No	Yes (on client_reference)	Purchase PIN
/v1/pins/{client_reference}	GET	Yes	Yes	Purchase status
/v1/checks	POST	No	No (consumes remaining)	Web check
/v1/verifications	POST	Yes	Yes	Verify (minimised)
/v1/schools/{centre_number}/results	GET	Yes	Yes	Bulk list
/v1/amendments	POST	No	Conditional	Record amendment

Filter vs identifier vs body (examiner favourite):

- **Identifier (path):** centre_number, client_reference — which resource.
- **Filter (query):** sitting on school bulk — which slice of a collection.
- **Body:** PIN value, subject list, amendment evidence — content of the operation.

Versioning: /v1/.... When /v2 ships, /v1 remains until a published deprecation date (minimum 12 months). A school still calling v1 gets v1 behaviour. Breaking changes only in a new major version.

Status codes (illustrative — full set is in the YAML):

- 201 PIN issued
- 202 purchase still in progress (same client_reference)
- 200 check / verify / bulk success
- 403 PIN exhausted / not authorised for centre
- 404 not released / unknown reference
- 402 payment declined
- 401 missing/invalid auth
- 409 concurrent amendment conflict
- 429 rate limit
- 503 partner unavailable (retry with **same** client_reference)

Machine-readable code in error body (e.g. PIN_EXHAUSTED) so tests and clients do not parse English messages (FR-069).

The full OpenAPI 3.0.3 document is in openapi-takarda-v1.yaml. Validate it with any OpenAPI validator before submission; fix paths if your final naming differs, but keep the properties above.

5.4 The PIN purchase, fixed

Plain language

The payment partner charges when the PIN is issued and does **not** accept an idempotency key. Last year that produced 9,100 double charges. We cannot change the partner before May. So we fix it **on our side** of the boundary.

Flow (happy path)

1. Client (agent, bank, or web) generates a client_reference (unique per attempt) and calls POST /v1/pins with that reference.
2. We look up client_reference:
 - **Never seen** → create row in state in_progress, call partner charge, on success create PIN (remaining = 5), set state issued, return PIN.
 - **Already issued** → return the **same** PIN and remaining count; **do not** charge again (FR-002).
 - **In progress** → return 202 / “still processing”; do not start a second charge (FR-003).
 - **Failed** → allow a **new** client_reference for a fresh attempt (or a defined retry policy).

State we keep (per client_reference)

- client_reference (unique key)
- status: in_progress | issued | failed
- pin (if issued)
- remaining_checks
- partner_charge_id (if partner returned one)
- amount_kobo = 350000
- timestamps

Money is always whole kobo.

Partner timeout — “unknown whether charge landed”

If the partner does not answer within our timeout:

1. Leave status as in_progress (or move to unknown if you prefer an explicit state).
2. **Do not** issue a PIN yet.
3. **Do not** invent a success.
4. Background reconciliation (below) or a timed follow-up query to the partner (if they offer status-by-reference) decides:
 - Charge succeeded → issue PIN, set issued, return PIN on next client poll (GET /pins/{client_reference}).
 - Charge failed / absent → set failed, client may retry with a **new** client_reference.

Client is told to poll status with the **same** client_reference, never to open a parallel purchase with a new reference while one is in progress.

Automatic detection and refund of double charges (no human spreadsheet)

1. **Daily (or more frequent) reconciliation job** reads the partner settlement file.
2. For each successful charge in the settlement file, find matching partner_charge_id / amount / time window in our ledger.
3. Rules:
 - Settlement success with **no** matching issued PIN → investigate / auto-issue or auto-refund per policy (prefer refund if PIN cannot be safely issued).
 - **Two** settlement successes that map to the same logical purchase attempt → mark duplicate; trigger **refund** of the extra charge via partner refund API or finance file that does not require a human to spot the row.
 - Issued PIN with **no** settlement success within T days → alert (our side or partner side mismatch).
4. Metrics: count of duplicates found, refunds issued, unmatched rows. Target aligns with BRD: **zero** double charges attributable to our layer.

ASSUMPTION: Partner provides a settlement file that includes enough fields (amount, timestamp, their transaction id, card/masked payer ref) to match our client_reference or partner_charge_id. If the file is weaker, matching quality drops and we state that limit explicitly.

6. Part D

6.1 Data model (plain language first)

What a “result” means

A released subject-result is a fact that must be reproducible as it stood on any past date. It is **never updated in place**. If a grade changes, we append an amendment that names the officer and the evidence. “Current grade” is derived: latest amendment for that subject, or the original released grade if there is none.

Core entities (conceptual)

Entity	Identity (key idea)	Important attributes	Notes
Candidate	Examination number (per sitting)	Names, DOB, centre, state	Stable identity for checks and verification
Sitting	Year + session (e.g. 2026-05)	Release flag, release timestamp	Noon flag lives here
SubjectResult	Candidate + sitting + subject	Original grade, released_at	Immutable after release
Amendment	New id; points at SubjectResult	Previous grade, new grade, officer id, evidence ref, amended_at	Append-only
Certificate	Certificate number	Links to candidate + sitting; issued metadata	Verification key
PIN	PIN value / id	client_reference, remaining_checks, status, amount_kobo	Money in whole kobo (350000)
CheckAudit	New id	PIN, channel, timestamp, remaining after	Audit only
VerificationAudit	New id	Verifier, subjects requested, grades returned, timestamp	Audit only
Centre	Centre number	School binding	Bulk authorisation

Cardinality (short)

- One candidate in one sitting has about nine SubjectResults.
- One SubjectResult has zero or more Amendments (ordered by time).
- One certificate maps to one candidate+sitting.
- One PIN has one remaining counter (0–5).

Money

Always whole kobo (integer). PIN price = 350000.

ASSUMPTION: Examination number is unique within a sitting; certificate number is unique enough to identify a certificate together with DOB for verification (as the brief’s query states).

6.2 Database architecture

Engine choice

A conventional relational engine with strong single-node transactions and mature B-tree indexes (e.g. PostgreSQL-class).

Why this workload, not a textbook default

- Writes are modest most of the year (<20 checks/s, ~475 amendments per sitting).
- The brutal load is a **read spike** of known shape for about one hour.
- We need “as of date” history and officer-named amendments — append-only rows fit relational tables cleanly.
- Six engineers cannot also operate a multi-shard distributed store before May.

Replication

- One primary for all writes (PIN, amendment, release load, audit).
- Read replicas for check, verification, and bulk reads that can tolerate short replica lag.
- Rule: anything that must see an amendment within the five-minute FR window after write is routed to primary (or to a replica only after lag is known to be below that threshold). Steady-state candidate checks may use replicas.

Partitioning (not full sharding in v1)

- Partition large fact tables by **sitting** (and optionally by centre within sitting for bulk).
- Why: almost every hot query is “for one sitting”. Old sittings are cold and retained for 50 years.
- **No candidate-ID sharding in v1** (ADR-07): cross-sitting regulator queries and “Maths by state” stay single-partition or few-partition operations.

What the partition key makes more expensive

A search that spans many sittings without a sitting filter (e.g. global surname search across 50 years) must touch many partitions. That is acceptable because that query is rare and internal (see Query 5).

6.3 Five queries — step by step what the engine does with the rows

For each query: **walk the rows first**, then indexes, access method, examined vs returned, growth, fix if needed, and what would prove you wrong.

Speed is mentioned only after the row story.

Query 1 — Candidate's nine results for one sitting, by examination number

What the engine does with the rows (before speed)

1. Locate the candidate row (or the subject-result rows) for this examination number and this sitting.
2. Read the nine subject-result rows for that candidate+sitting.
3. For each subject, find whether any amendments exist; if so, take the latest by time; otherwise use the original grade.
4. Return nine grade lines (and only those).

Index

- Leading: (sitting_id, examination_number) on the result side (or equivalent candidate key), with subject available for the nine rows.
- Amendments: (subject_result_id, amended_at) so "latest amendment" is a short range.

Access method

Index seek on sitting + examination number → retrieve ~9 result rows → nested lookup (or small join) to latest amendment per subject. Not a table scan.

Rows examined vs returned

Examined: on the order of 9 result rows + a handful of amendment rows if any. Returned: 9. Those numbers stay close.

Growth to 216M+ rows

Does **not** get slower in a meaningful way if the index leads with sitting (or the table is partitioned by sitting). The engine only touches one sitting's slice.

If too slow

You almost never need a fix here; if you do, materialise "current grade" per subject at amendment time (extra write cost on the rare amendment path).

What would prove me wrong

Observation: access method is a sequential scan of the whole results table, or rows examined are orders of magnitude above ~10–20 for a single candidate.

One plain sentence you can say out loud

"This is a point lookup of nine rows in one sitting; a good index makes the engine touch only those rows, and archive growth does not matter."

Query 2 — School bulk: every result for about 400 candidates in one sitting

What the engine does with the rows

1. Restrict to one sitting and one centre_number.
2. Find all candidates registered under that centre for that sitting (~400).
3. For each, read their ~9 subject results and resolve current grade (original + latest amendment).
4. Build the bulk payload (~3,600 grade lines).

Index

- (sitting_id, centre_number, examination_number) on candidates or results so the centre slice is contiguous.
- Same amendment lookup as Query 1.

Access method

Index range scan on sitting + centre → ~400 candidates → ~3,600 result rows → amendment lookups. Not a full table scan.

Rows examined vs returned

Examined ≈ returned order of magnitude (thousands, not millions). Examined and returned stay close.

Growth to 216M

Stable if partitioned/indexed by sitting: only one sitting's centre slice is read.

If too slow

Pre-build bulk files per centre at release time (same idea as the surge pre-generation path); cost is generation time before noon, not query time at noon.

What would prove me wrong

Observation: the plan scans all candidates in the sitting (hundreds of thousands) and filters centre late, or examined rows are ~1.9M for a 400-candidate school.

One plain sentence

“The engine should walk only one centre’s few thousand rows in one sitting, not the whole cohort.”

Query 3 — Employer verifies one certificate by certificate number + date of birth (DPO-minimised)

What the engine does with the rows

1. Find the certificate by certificate number.
2. Confirm DOB matches the linked candidate (authentication of the request, not a data dump).
3. Read only the subject grades the caller asked for (or the certificate’s scoped subjects).
4. Resolve current grade via amendments.
5. Return validity + those grades only — never other subjects, address, phone, or centre.

Index

- Unique/primary style lookup on certificate_number.
- Candidate key for DOB check.
- Subject results by candidate+sitting+subject; amendments as above.

Access method

Index seek on certificate number → few rows → seek subject results for the requested subjects only. Tight.

Rows examined vs returned

Examined: handful. Returned: handful (only requested subjects). Close together.

Growth to 216M

Irrelevant to this path if certificate number is selectively indexed; one certificate does not care how many others exist.

If too slow

Almost never; if certificate table is huge and unindexed, add the certificate_number index (cheap selective lookup).

What would prove me wrong

Observation: response contains subjects not requested, or plan shows a scan of all results for the candidate when only one subject was asked for.

One plain sentence

“This is a single-certificate point lookup that must return only the fields the DPO allows — the engine should touch a handful of rows and stop.”

Query 4 — Count how many candidates received each grade in Mathematics, by state, for one sitting

(Every candidate sits Mathematics.)

What the engine does with the rows

1. Restrict to one sitting.
2. Restrict to subject = Mathematics.
3. For each Mathematics result in that sitting, resolve **current** grade (original or latest amendment).
4. Group by state (from candidate/centre) and by grade; count.

This is an **aggregation over essentially the whole sitting’s Maths results** (~1.9M rows for the main sitting), not a point lookup.

Index

- (sitting_id, subject_code) on results helps the engine find Maths rows in that sitting without scanning other subjects.
- State may sit on candidate; join candidate ↔ result.
- Amendments still needed for “current” grade — that can force extra work per row if many amendments exist (they do not; ~475/sitting total).

Access method

Index range scan (or partition scan) of all Mathematics rows for the sitting, join to candidate for state, aggregate. This is **not** a tiny seek. A covering structure (sitting, subject, state, grade) can reduce heap visits, but the engine still **must visit on the order of 1.9M Maths facts** to count them.

Rows examined vs returned

Examined: ~1.9M (main sitting). Returned: a small grid (states × grade bands) — dozens of rows. **Examined >> returned**. That gap is the signal.

Growth to 216M

As archive grows, **this query does not get slower** if it is always constrained to **one sitting** and the table is partitioned by sitting — it only ever scans that sitting’s Maths slice. If someone runs it without a sitting filter, it becomes catastrophic; the API must require sitting.

If too slow

Maintain a small summary table per sitting (state × grade counts) updated at release and on each Maths amendment. Cost: extra write on amendment and at release; report becomes a tiny read. Trade-off: summary can drift if amendment path forgets to update it — must be in the same transaction as the amendment.

What would prove me wrong

Observation: plan shows a full scan of all 216M rows, or examined rows $\approx$ 216M when sitting was supplied; or latency stays high after a per-sitting summary exists.

One plain sentence

“This has to look at about 1.9 million Mathematics rows for that sitting and collapse them into a small grid — examined rows will dwarf returned rows, and that is expected.”

Query 5 — Officer search: partial surname, match anywhere in the name

(The one where a normal index often does NOT help; and the one that does not need to be fast.)

What the engine does with the rows

1. Take a substring typed by the officer (e.g. “ada”).
2. Find candidates whose name **contains** that substring anywhere (not only at the start).
3. Return matching candidates for the officer to pick from (internal tool).

Why a normal B-tree index on surname does NOT help

A B-tree on surname supports “starts with” (ada%) efficiently. It does **not** support “contains” (%ada%) — the engine cannot binary-search into the middle of the string. So with only a plain B-tree, the engine falls back to **scanning large numbers of candidate rows** and filtering by substring.

Access method (with only ordinary indexes)

Sequential scan or large range scan of the candidate table (or of a sitting’s candidates if filtered), apply “name contains substring” row by row. Examined rows $\gg$ returned rows.

Optional specialised index

A trigram / full-text style index *can* help “contains” searches — but it costs storage and write overhead on every name change, and this query is rare.

Rows examined vs returned

Without a trigram index: examined could be hundreds of thousands to millions; returned maybe tens. With a sitting filter, examined drops to that sitting’s candidates (~1.9M max still large). Returned stays small.

Growth to 216M

Gets worse if the search is global across all history with no sitting filter. With a required sitting filter it is bounded by one sitting’s candidates.

Does this query need to be fast?

No.

It is an internal officer tool, used occasionally when someone remembers only part of a name. It is not on the results-day path, not on the verification path, and not on the PIN path. Making it ultra-fast is optional optimisation, not a May success criterion.

If you “fix” it

Add a trigram (or similar) index on name — cost is write amplification and storage on the candidate table; benefit is only for this rare internal search. Prefer requiring sitting (and maybe centre) in the UI before search, so the scan is bounded.

What would prove me wrong

Observation: this query appears on the critical results-day path, or product metrics show officers running unscoped global contains-searches thousands of times per hour.

One plain sentence you can say out loud

“A normal index cannot help a ‘contains anywhere in the name’ search, so the engine may scan a lot of rows — and that is acceptable because this is a rare internal lookup, not a results-day path.”

Which two “catch people who only know indexes are fast”

1. **Query 4** — people say “index on subject” and expect a seek; in reality the engine must still visit ~1.9M Maths rows to count them. Examined $\gg$ returned is normal.
2. **Query 5** — people add a B-tree on surname and expect magic; “contains anywhere” does not use that B-tree. The honest answer is scan (or a specialised index you may choose not to pay for).

The query that does not need to be fast: Query 5 — one plain sentence above.

6.4 What every index costs + one you refuse

Index (conceptual)	Write cost	Storage	Effect when loading 1.9M results pre-release
(sitting, examination_number) on results	Extra maintain on each result insert	Moderate per sitting	Slightly slower bulk load; needed for Query 1
(sitting, centre_number) on candidates/results	Same	Moderate	Needed for Query 2 / bulk

certificate_number	Low (one row per certificate)	Low	Needed for Query 3
(sitting, subject_code) on results	Extra on insert	Moderate	Helps Query 4 find Maths slice; still scans that slice
Amendment (subject_result_id, amended_at)	Extra on each of ~475 amendments	Tiny	Negligible at release load
Trigram on name (Query 5)	High on every candidate write/update	High	Hurts release load and steady writes for a rare query

Index I choose **NOT** to create (for v1)

The trigram (or equivalent “contains”) index on candidate name.

Why: Query 5 does not need to be fast; the write and storage cost hit every candidate insert and the 1.9M load; six engineers gain little for May. Revisit if officers run this constantly without a sitting scope (reverse signal).

6.5 “What would prove me wrong” — collected thresholds

Query Observation that falsifies the prediction

- 1 Sequential scan of results, or examined rows $\gg$ 20 for one candidate
- 2 Examined rows on order of full sitting (~1.9M) for a 400-candidate centre
- 3 Forbidden fields in response, or scan of all subjects when one was requested
- 4 Full archive scan when sitting was provided; or examined $\approx$ 216M
- 5 This query becomes hot path on results day, or unscoped global contains runs at high QPS

7. Part E

AI was used throughout design (requirements wording, architecture options, OpenAPI sketch, query reasoning). The brief requires at least one case where AI output was checked, found wrong, and corrected. That case is recorded first.

Mandatory correction (AI was wrong)

Field	Detail
Date	2026-09-28
Tool	Grok (architecture coaching session)
Task	Draft OpenAPI 3.0.3 security scheme for employers' association mTLS
AI claim	Used type: mutualTLS under components.securitySchemes
How checked	Ran npx @redocly/cli lint api/openapi.yaml
Finding	Error: OpenAPI 3.0.3 allows only apiKey, http, oauth2, openIdConnect. mutualTLS is valid in OpenAPI 3.1, not 3.0.3. Spec failed validation.
What was actually true	For a 3.0.3 contract, mTLS must be described as edge/transport policy (and optionally represented with a non-normative scheme or extension), not as type: mutualTLS.
Decision	Removed invalid mutualTLS type. Documented association mTLS in the API description and info block as edge-enforced client certificates . Kept Bearer for human verifiers/schools. Spec now validates. Design intent (machine ≠ person auth) unchanged.

This is the required “AI was confident and incorrect; human verification changed the artefact” entry.

Other AI-assisted tasks (summary)

DATE	TOOL	TASK	OUTCOME
2026-09-25	Grok	BRD / PRD / FRD / VBS drafts	Accepted structure; numbers taken only from brief; edited for board vs builder vs tester voice
2026-09-28	Grok	Conflict list and impossible request	Accepted: Ops “rewrite already-verified certificates” marked impossible; append-only + future-read consistency offered instead
2026-09-28	Grok	C4 Mermaid then draw.io	Diagrams retained; “deliberately left out” paragraphs added per brief
2026-09-28	Grok	Five query predictions	Row-first walkthroughs retained; Query 5 “contains” limitation and “need not be fast” kept after cross-check with B-tree behaviour
2026-09-29	Grok	OpenAPI lint fix	See mandatory correction above

Method

- Every number in the dossier is from the brief, calculated from the brief, or labelled ESTIMATE/ASSUMPTION.
- AI suggestions that introduced generic reasons (“microservices are complex”) were rejected; only Takarda-specific rejection properties were kept.
- No application or database was built; AI was not used to generate runnable code for submission.